



SEMESTER END EXAMINATIONS FEBRUARY - MARCH 2021

Program	: B.E.: Information Science and Engineering	Semester	: III
Course Name	: Data Structures	Max. Marks	: 100
Course Code	: IS35	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit

UNIT- I

- Discuss briefly the history of file structure design. CO1 (10)
 - Compare Data structures and File structures. CO1 (05)
 - With an example, show that the factorial of a number can be computed using recursion. CO1 (05)
- Differentiate between physical files and logical files. CO1 (05)
 - Define Recursion. Design a recursive program to solve the towers of Hanoi problem using C/C++. Give the trace for 3 discs. CO1 (10)
 - What are the different ways of initializing one and two dimensional arrays? CO1 (05)

UNIT – II

- Discuss the doubly linked lists with advantages and disadvantages. Write a C function to delete a node from a circular doubly linked list with header node. CO2 (10)
 - What is a linked list? Illustrate the different types of linked lists with neat diagram. CO2 (10)
- What is circular linked list? Write a C program to implement search function and delete a node given its position in a circular linked list. CO2 (10)
 - Implement the following using singly linked list: CO2 (10)
 - Count the number of nodes in the list
 - Reverse the list.

UNIT – III

- Define stack. List and implement basic operations on stack. CO3 (10)
 - What are priority queues? Design a C code to implement priority queues insertion operation using arrays. CO3 (06)
 - What are the major drawbacks of ordinary queues? How do you overcome them? CO3 (04)
- Design a C code to convert an expression from infix to postfix form using stack. CO3 (08)
 - Explain the working principle of circular queue with neat diagrams. CO3 (05)
 - Design an algorithm to evaluate a postfix expression using stack and evaluate the expression $623+ - 382/+ * 2^3+$ using the same. CO3 (07)

UNIT – IV

- Write a program that implements a queue using linked lists. Test the program with five integers. CO4 (08)
 - Discuss how binary trees are represented using lists. Mention the drawbacks of representing binary trees using arrays. CO4 (08)
 - Justify how linked lists are better than array data structures. CO4 (04)

8. a) Write an algorithm to perform recursive search and iterative search for a given element in a Binary Search Tree. CO4 (10)
b) Write routine for CO4 (10)
i. create a binary tree ii. In order tree traversal of a Binary Tree.

UNIT – V

9. a) Give any two points each to compare and contrast B-Trees and B+ Trees. CO5 (12)
With any example values of your choice illustrate and explain the step-by-step logic used for the following:
i. Creation of a B+ Tree out of those values
ii. Inserting a new value into the B+ Tree.
b) What is Hashing? Why do you need it? Explain a Simple Hashing function algorithm. CO5 (08)
10. a) Define B-Tree. Explain with an example the algorithm for inserting an element into this B-Tree. CO5 (10)
b) What is 'Collision'? Explain *any two* Collision Handling techniques. CO5 (10)
